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itm-data2

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1

Python

spark.sql("SHOW VOLUMES")

DataFrame(database: string, volume_name: string)

See performance (1)

Optimize

2: Cell 1

dbutils.fs.mkdirs("/Volumes/itm_data2/itm_piyush/raw/lakehouse_lab3")

dbutils.fs.mkdirs("/Volumes/itm_data2/itm_piyush/raw/lakehouse_lab3/bronze")

dbutils.fs.mkdirs("/Volumes/itm_data2/itm_piyush/raw/lakehouse_lab3/silver")

dbutils.fs.mkdirs("/Volumes/itm_data2/itm_piyush/raw/lakehouse_lab3/gold")

See performance (4)

Optimize

3: Cell 2

dbutils.fs.ls("/Volumes/itm_data2/itm_piyush/raw/lakehouse_lab3")

See performance (1)

Optimize

4

import urllib.request

import json

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4

Python

import urllib.request

import json

List CSV files from Github repo

api_url = "https://api.github.com/repos/databricks/Spark-The-Definitive-Guide/contents/data/retail-data/all"

with urllib.request.urlopen(api_url) as response:

files = json.loads(response.read()).decode()

Download each CSV to the volume

csv_dir = "/Volumes/itm_data2/itm_piyush/raw/lakehouse_lab3/source_csv"

dbutils.fs.mkdirs(csv_dir)

for f in files:

if f["name"].endswith(".csv"):

urllib.request.urlretrieve(f["download_url"], f"{csv_dir}/{f['name']}")

Read all CSVs with Spark

df_raw = spark.read.option("inferSchema", True).option("header", True).csv(f"{csv_dir}/*.csv")

See performance (1)

Optimize

5

display(df_raw)

See performance (1)

Optimize

Table

+

InvoiceNo StockCode Description Quantity InvoiceDate UnitPrice CustomerID Country

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> df_raw: pyspark.sql.connect.dataframe.DataFrame = [InvoiceNo: string, StockCode: string ... 6 more fields]

See performance (1)

Optimize

5

display(df_raw)

See performance (1)

Optimize

Table

+

InvoiceNo StockCode Description Quantity InvoiceDate UnitPrice CustomerID Country

1	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850	United Kingdom
2	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850	United Kingdom
3	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850	United Kingdom
4	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850	United Kingdom
5	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850	United Kingdom
6	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	12/1/2010 8:26	7.65	17850	United Kingdom
7	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	12/1/2010 8:26	4.25	17850	United Kingdom
8	536366	22633	HAND WARMER UNION JACK	6	12/1/2010 8:28	1.85	17850	United Kingdom
9	536366	22632	HAND WARMER RED POLKA DOT	6	12/1/2010 8:28	1.85	17850	United Kingdom
10	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	12/1/2010 8:34	1.69	13047	United Kingdom
11	536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	12/1/2010 8:34	2.1	13047	United Kingdom
12	536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	12/1/2010 8:34	2.1	13047	United Kingdom
13	536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	12/1/2010 8:34	3.75	13047	United Kingdom
14	536367	22310	IVORY KNITTED MUG COSY	6	12/1/2010 8:34	1.65	13047	United Kingdom
15	536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	12/1/2010 8:34	4.25	13047	United Kingdom

10,000+ rows | Truncated data | 1.824s runtime

Refreshed 33 minutes ago

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6: Cell 5 Python

```
df_raw.write.format("delta").mode("overwrite").save("/Volumes/ltm_data2/ltm_piyush/raw/lakehouse_lab3/bronze/retail")
```

See performance (1) Optimize

7: Cell 6

```
spark.sql("""
CREATE OR REPLACE TABLE bronze_retail
AS SELECT * FROM delta.`/Volumes/ltm_data2/ltm_piyush/raw/lakehouse_lab3/bronze/retail`
""")
```

See performance (1) Optimize

DataFrame[num_affected_rows: bigint, num_inserted_rows: bigint]

8

```
spark.sql("SELECT * FROM bronze_retail LIMIT 10")
```

DataFrame[InvoiceNo: string, StockCode: string, Description: string, Quantity: int, InvoiceDate: string, UnitPrice: double, CustomerID: int, Country: string]

9

```
df_silver = df_raw.dropna()
```

df_silver: pyspark.sql.connect.dataframe.DataFrame = [InvoiceNo: string, StockCode: string ... 6 more fields]

10

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10

```
df_silver = df_silver.withColumnRenamed("InvoiceNo", "InvoiceNo"), withColumnRenamed("Description", "description")
```

df_silver: pyspark.sql.connect.dataframe.DataFrame = [InvoiceNo: string, StockCode: string ... 6 more fields]

11

```
from pyspark.sql.functions import col
df_silver = df_silver.filter(col("Quantity") > 0)
```

df_silver: pyspark.sql.connect.dataframe.DataFrame = [InvoiceNo: string, StockCode: string ... 6 more fields]

12: Cell 11

```
df_silver.write.format("delta").mode("overwrite").save("/Volumes/ltm_data2/ltm_piyush/raw/lakehouse_lab3/silver/retail_clean")
```

See performance (1) Optimize

13: Cell 12

```
%sql
CREATE OR REPLACE TABLE silver_retail
AS SELECT * FROM delta.`/Volumes/ltm_data2/ltm_piyush/raw/lakehouse_lab3/silver/retail_clean`;
```

See performance (1) Optimize

Table +

num_affected_rows num_inserted_rows

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13: Cell 12

```
%sql
CREATE OR REPLACE TABLE silver_retail
AS SELECT * FROM delta.`/Volumes/ltm_data2/ltm_piyush/raw/lakehouse_lab3/silver/retail_clean`;
```

See performance (1) Optimize

Table +

num_affected_rows num_inserted_rows

No rows returned

0 rows | 4.124s runtime Refreshed 30 minutes ago

This result is stored as _sqldf and can be used in other Python and SQL cells. Send feedback

14

```
%sql
SELECT COUNT(*) FROM silver_retail;
```

See performance (1) Optimize

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11:24 AM (2s) 14

SQL

```
SELECT COUNT(*) FROM silver_retail;
```

See performance (1) Optimize

COUNT(*)
397924

1 row | 1.531s runtime Refreshed 31 minutes ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

11:25 AM (3s) 15: Cell 14

```
from pyspark.sql.functions import sum, col, to_date

df_gold = df_silver.withColumn("date", to_date("InvoiceDate", "M/d/yyyy H:mm")) \
    .groupBy("date") \
    .agg(sum(col("Quantity")) * col("UnitPrice")).alias("daily_revenue")
```

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11:25 AM (3s) 15: Cell 14

```
from pyspark.sql.functions import sum, col, to_date

df_gold = df_silver.withColumn("date", to_date("InvoiceDate", "M/d/yyyy H:mm")) \
    .groupBy("date") \
    .agg(sum(col("Quantity")) * col("UnitPrice")).alias("daily_revenue")

df_gold.write.format("delta").mode("overwrite").save("/Volumes/ltm_data2/ltm_piyush/raw/lakehouse_lab3/gold/daily_revenue")

df_gold: pyspark.sql.connect.dataframe.DataFrame = [date: date, daily_revenue: double]
```

See performance (1) Optimize

11:25 AM (3s) 16: Cell 15

```
SQL
CREATE OR REPLACE TABLE gold_daily_revenue
AS SELECT * FROM delta.`/Volumes/ltm_data2/ltm_piyush/raw/lakehouse_lab3/gold/daily_revenue`;
```

See performance (1) Optimize

num_affected_rows	num_inserted_rows
-------------------	-------------------

No rows returned

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df_gold: pyspark.sql.connect.dataframe.DataFrame = [date: date, daily_revenue: double]

11:25 AM (3s) 16: Cell 15

```
SQL
CREATE OR REPLACE TABLE gold_daily_revenue
AS SELECT * FROM delta.`/Volumes/ltm_data2/ltm_piyush/raw/lakehouse_lab3/gold/daily_revenue`;
```

See performance (1) Optimize

num_affected_rows	num_inserted_rows
-------------------	-------------------

No rows returned

0 rows | 3.325s runtime Refreshed 30 minutes ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

11:25 AM (2s) 17

```
SQL
SELECT * FROM gold_daily_revenue ORDER BY date;
```

See performance (1) Optimize

df_gold: pyspark.sql.connect.dataframe.DataFrame = [date: date, daily_revenue: double]

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Run all Serverless Schedule Share

11:26 AM (2s) 17

%sql

SELECT * FROM gold_daily_revenue ORDER BY date;

See performance (1)

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [date: date, daily_revenue: double]

	date	daily_revenue
91	2011-03-27	9224.399999999999
92	2011-03-28	20005.159999999993
93	2011-03-29	37190.519999999993
94	2011-03-30	29587.420000000003
95	2011-03-31	25688.610000000006
96	2011-04-01	24586.820000000005
97	2011-04-03	6918.499999999998
98	2011-04-04	29017.770000000002
99	2011-04-05	28759.370000000004
100	2011-04-06	12722.320000000002
101	2011-04-07	16827.770000000003
102	2011-04-08	22720.710000000003
103	2011-04-10	9913.979999999997
104	2011-04-11	20184.140000000003
105	2011-04-12	24023.960000000005

305 rows | 1.517s runtime

Refreshed 31 minutes ago

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